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A home-monitoring system for sleep disorders

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1. Context Analysis

1.1. Clinical Background

Sleep disorders are highly prevalent and often under-recognized conditions that can significantly affect both physical and mental health. Nearly one in two individuals will experience sleep disturbances at some point in their life, yet sleep-related issues are often only briefly addressed in standard medical education(Guillodo et al., 2020). Poor sleep quality has been associated with a wide range of adverse outcomes, including increased risk for cardiovascular diseases, impaired cognitive performance, mood disorders, and reduced quality of life(K. Pavlova & Latreille, 2019).

In recent years, wearable devices have emerged as promising tools for ecological sleep monitoring. Unlike traditional polysomnography (PSG), which is limited to single-night, clinical-lab settings, wearable activity trackers (WATs) allow for multi-night recordings in naturalistic environments. These devices are generally comfortable, easy to use, and minimally invasive, making them suitable for home-based monitoring. However, their accuracy can vary. Many rely on accelerometers to infer sleep patterns, which may struggle to detect sleep latency or differentiate between quiet wakefulness and light sleep. Others incorporate heart rate and heart rate variability (HRV) to identify sleep stages, but current algorithms still face challenges in reliably classifying these stages (Guillodo et al., 2020).

Despite these limitations, advances in sensor miniaturization and data analytics are rapidly improving the performance of wearables. Integration of wearable data into digital health applications offers new possibilities for personalized care and long-term sleep monitoring (Lloyd et al., 2025).

Regarding clinical management, insomnia and other sleep disorders are often treated through a combination of behavioral and pharmacological interventions. The Cognitive Behavioral Therapy for Insomnia (CBT-I) is strongly recommended by the American Academy of Sleep Medicine as a first-line treatment. Nonetheless, pharmacological therapy remains widely used, especially for short-term relief or when CBT-I is not available (Han et al., 2012).

Pharmacological options include orexin receptor antagonists, benzodiazepines, non-benzodiazepine hypnotics, melatonin receptor agonists, and certain antidepressants. Some of these medications, particularly benzodiazepines, carry a risk of dependence, withdrawal, and cognitive impairment, especially when used long-term. Therefore, it is clinically relevant to track the duration of each prescription (K. Pavlova & Latreille, 2019).

1.2. Aim of the project

The aim of this project is to develop an intuitive and clinically useful interface designed to support both general practitioners and patients in the long-term management and monitoring of sleep disorders. The system is intended for patients who have been diagnosed with sleep-related conditions or for those at risk of relapse, offering a reliable way to track sleep patterns through wearable devices in real-world, everyday settings.

The interface integrates multiple components to provide a comprehensive overview of the patient's condition. It collects and displays data from wearable devices — which monitor sleep quality and related parameters — alongside standardized clinical questionnaires that assess the patient's subjective experience and mental health status. This dual approach aims to combine objective measurements with patient-reported outcomes to better guide clinical decisions.

Additionally, the platform features a prescription tracker to help healthcare providers monitor the use of pharmacological treatments, which is especially relevant in the context of sleep disorders, as several commonly prescribed medications carry a risk of dependence and withdrawal symptoms when used long-term. Displaying prescription timelines directly in the patient's profile can support safer and more informed prescribing practices.

Lastly, the interface also serves as a scheduling tool for managing medical appointments — not only with the general practitioner, but also for specialist referrals, which are typically initiated by the primary care provider in many healthcare systems. This integrated system is designed to streamline communication between patients and healthcare professionals, support continuity of care, and encourage a more proactive approach to sleep health management.

1.3. Main literature findings

Article	Key concepts
Yuan, W., Chen, L., Wu, Y., Su, B., Liu, J., Zhang, Y., Chen, M., Ma, Y., Guo, T., Wang, X., Ma, T., Ma, Q., Cui, M., Ma, J., & Dong, Y. (2023). **Sleep time and quality associated with depression and social anxiety among children and adolescents aged 6–18 years, stratified by body composition.** *Journal of Affective Disorders*, *338*, 321–328. https://doi.org/10.1016/j.jad.2023.06.029	The article highlights that poor sleep quality or short sleep duration is linked to increased depression and social anxiety in children and adolescents. It also shows that body composition plays a moderating role: higher muscle mass and lower body fat can reduce the negative impact of poor sleep on mental health. The findings suggest that both sleep habits and physical fitness should be addressed together when aiming to prevent or manage mental health issues in youth.
Hirshkowitz, M., Whiton, K., Albert, S. M., Alessi, C., Bruni, O., DonCarlos, L., Hazen, N., Herman, J., Adams Hillard, P. J., Katz, E. S., Kheirandish-Gozal, L., Neubauer, D. N., O'Donnell, A. E., Ohayon, M., Peever, J., Rawding, R., Sachdeva, R. C., Setters, B., Vitiello, M. v., & Ware, J. C. (2015). **National Sleep Foundation's updated sleep duration recommendations: Final report.** *Sleep Health*, *1*(4), 233–243. https://doi.org/10.1016/j.sleh.2015.10.004	This final report from the National Sleep Foundation provides updated sleep duration recommendations across all age groups, from infants to older adults. A multidisciplinary panel of experts reviewed recent scientific evidence through a structured consensus process to define recommended, potentially appropriate, and not recommended sleep durations. The goal is to offer clear, evidence-based guidelines to support health and well-being through adequate sleep across the lifespan.
K. Pavlova, M., & Latreille, V. (2019). **Sleep Disorders**. In *American Journal of Medicine* (Vol. 132, Issue 3, pp. 292–299). Elsevier Inc. https://doi.org/10.1016/j.amjmed.2018.09.021	This review article focuses on the diagnosis and management of six major categories of sleep disorders. It provides an overview of key features of insomnia, circadian rhythm disorders, sleep-related breathing disorders, hypersomnia/narcolepsy, parasomnias, and restless legs syndrome/periodic limb movement disorder. The primary goal is to assist clinicians in effectively screening for and treating these common conditions that significantly affect patients' health and quality of life.
Guillodo, E., Lemey, C., Simonnet, M., Walter, M., Baca-García, E., Masetti, V., Moga, S., Larsen, M., Ropars, J., & Berrouiguet, S. (2020). **Clinical applications of mobile health wearable–based sleep monitoring: Systematic review**. In *JMIR mHealth and uHealth* (Vol. 8, Issue 4). JMIR Publications Inc. https://doi.org/10.2196/10733	This systematic review explores the clinical use of mobile wearable devices for sleep monitoring. The authors evaluated multiple studies to assess the feasibility and reliability of these devices compared to polysomnography, the clinical gold standard. While wearables show promise for real-world sleep tracking, the review highlights significant limitations in their accuracy. The article emphasizes the need for

	further research to enhance their precision and long-term clinical use.
Han, K. S., Kim, L., & Shim, I. (2012**). Stress and Sleep Disorder**. *Experimental Neurobiology*, *21*(4), 141–150. https://doi.org/10.5607/en.2012.21.4.141	This scientific review explores the strong bidirectional link between stress and sleep disorders, focusing on the physiological, hormonal, and neural mechanisms involved. It highlights the central role of the hypothalamic-pituitary-adrenal (HPA) axis in stress response and its regulation during sleep. The article discusses how stress can trigger insomnia and how it may become chronic through predisposing, precipitating, and perpetuating factors. It also addresses the involvement of the immune system and circadian rhythms, reviews theoretical models of insomnia, and suggests daily stress management strategies as a preventive and therapeutic approach to sleep disturbances.
Lloyd, R. M., Crawford, T., Donald, R., Gray, D. D., Healy, W. J., Junna, M. R., Lewin, D., Revana, A., & Schutte-Rodin, S. (2025). **Quality measures for the care of patients with insomnia: 2024 update after measure maintenance.** *Journal of Clinical Sleep Medicine*, *21*(3), 565–571. https://doi.org/10.5664/jcsm.11512	This scientific article presents a 2024 update to the quality measures for insomnia care, originally published by the American Academy of Sleep Medicine in 2015. A dedicated task force revised these measures to align with current medical evidence and clinical guidelines. The update aims to provide tools for evaluating and improving the management of insomnia in both adults and children, using process and outcome indicators focused on sleep satisfaction, daytime functioning, and assessment of treatment side effects.

1.4. Introduction to our system

This system is designed to support the management and monitoring of sleep disorders by integrating input from three key actors: the patient, the general practitioner, and a wearable device. It provides a centralized platform where clinical data, wearable-tracked sleep parameters, and patient-reported outcomes can be accessed and managed to facilitate more informed and personalized healthcare decisions.

1.4.1. Patient profile

A new patient is registered in the system after being diagnosed with a sleep disorder or when placed under observation for suspected sleep disturbances or relapse risk. The initial patient profile is created by entering basic personal information: Name, Surname, Tax Code, Place of Birth, Date of Birth, Address, Phone Number, and E-mail.

During the first medical consultation, the physician completes the patient's profile by adding relevant clinical information, including any existing diagnoses, therapy plans, and other health-related data. At this stage, the patient is also provided with login credentials to access the system independently.

Once access is granted, the patient can review upcoming appointments, prescription plans, and clinical records, as well as actively participate in the data collection process. Patients are equipped with a wearable device that continuously records sleep-related metrics, allowing long-term, real-world monitoring of their sleep patterns. Additionally, patients are periodically asked to complete standardized questionnaires designed to assess sleep quality, mental health status, and lifestyle factors.

The system also allows patients to request medical appointments directly through the interface, whether for routine check-ups or to report the recurrence or worsening of symptoms. Appointment requests can include general consultations with the primary care physician or referrals to sleep specialists, depending on clinical needs.
Patients may be assigned to more than one physician, allowing collaboration between general practitioners and specialists to ensure integrated and continuous care.

1.4.2. Doctor profile

The doctor is the primary healthcare professional responsible for the clinical management and follow-up of patients affected by sleep disorders or at risk of sleep-related issues. Each doctor has a dedicated profile containing personal and professional information, including Name, Surname, Medical License Number, Specialization, and Contact Details.
Upon the patient's first consultation, the doctor is responsible for completing the patient's clinical profile by entering all relevant health information, such as diagnoses, therapy plans, and ongoing treatments. The doctor also defines the monitoring strategy, including the expected frequency of questionnaire completion and the use of wearable devices for sleep tracking.
The system allows the doctor to access and review real-time sleep data collected by the wearable devices, as well as the results of standardized questionnaires completed by the patient. This integrated information supports clinical decision-making, enabling early detection of relapses or deteriorations in the patient's condition.
In addition to managing clinical records, the doctor can also handle medication prescriptions directly through the system. A dedicated feature tracks the duration of pharmacological treatments, which is particularly relevant for medications that may cause dependence or withdrawal effects.
The system also includes an integrated appointment management tool. Doctors can schedule follow-up visits and respond to appointment requests submitted by patients, either for routine monitoring or for the evaluation of new or recurring symptoms. Furthermore, the platform allows the doctor to forward specialist referrals, ensuring seamless care coordination.
The doctor's interface is designed to offer a clear and organized view of all assigned patients, supporting continuous and personalized patient care.

1.4.3. Wearable Device Profile

The wearable device represents a key component in the continuous monitoring of sleep-related parameters in patients affected by sleep disorders or under observation for suspected disturbances. Each device is linked to an individual patient profile and is responsible for collecting objective physiological data in real-life conditions over extended periods.
The device is designed to monitor various sleep-related metrics, such as sleep duration, sleep stages, wake periods, and, in some cases, physiological signals like heart rate and heart rate variability. These data provide the physician with valuable insights into the patient's sleep quality and allow the early identification of irregularities or potential relapses.
Data collected by the wearable device are automatically synchronized with the system, ensuring that physicians have timely access to updated information for clinical evaluation. This process enables longitudinal monitoring, complementing the self-reported data gathered through standardized questionnaires.
Wearable devices are selected based on their usability, comfort, and ability to minimize interference with the patient's natural sleep. They are intended to empower both patients and physicians with a reliable, non-invasive method for assessing sleep patterns outside of laboratory settings.
The integration of wearable data into the system enhances the overall clinical picture, facilitating more accurate and personalized treatment strategies, and supporting proactive healthcare management.

2.Textual descriptions

The following section is dedicated to the textual descriptions we have implemented to model our system. Each description details a functionality of the system, accessible through the user's interface.

2.1. User

Doctor and patient can perform the following operations

2.1.1. Create new account

Title: Create new account
Pre-condition: the user has downloaded the app and has selected "Create new account"
Post-condition: the user has created his account
Main success scenario 1. The system asks if the user is a patient or a doctor. 2. The user selects "patient's account" or "doctor's account". 3. The system requires SPID/CIE verification. 4. The user is authenticated through SPID/CIE system. 5. The system checks SPID/CIE verification. 6. The system asks for mandatory information (patient: name, surname, fiscal code, date of birth, place of residence, height, weight, e-mail, phone number, doctor ID; doctor: name, surname, fiscal code, office address, phone number, e-mail). 7. The user inserts mandatory information. 8. The system checks if all data are correct. 9. The system sends an e-mail with a verification code. 10. The system asks to insert the verification code. 11. The user reads the email and inserts the verification code. 12. The system displays another window with username and password. 13. The user inserts his email as username. 14. The user inserts a password. 15. The user inserts the same password (check purpose). 16. The system checks if email and password are correct. 17. The system creates a new account and updates the database. 18. The system displays a confirmation message.
Alternative scenario: 6a. **Incorrect SPID/CIE** 1. The system detects an incorrect identification and the activity ends. 9a. **Empty field** 1. The system detects an empty field. 2. The system displays an error message, return to MSS at step 1. 9b. **User already registered**

1. The system detects the same info in the database.
2. The system displays an error message and the activity ends.

12a. **Wrong code for less than 3 attempts**

1. The system detects a worn verification code for less than 3 tries, return to MSS at step 5.

12b. **Wrong code for more than 3 attempts**

1. The system detects a worn verification code for more than 3 tries.
2. The system displays an error message, return to MSS at step 4.

17a. **Incorrect e-mail or password**

1. The system detects an email different from the previous one or a difference in the two inserted passwords.
2. The system displays an error message, return to MSS at step 8.

2.1.2. Login

Title: Login
Pre-condition: the user has opened the app and has selected "Login"
Post-condition: the user is logged in the app
Main success scenario 1. The system asks for username and password. 2. The user inserts his e-mail as username. 3. The user inserts the password. 4. The system checks if username and password are correct. 5. The system displays a confirmation message.
Alternative scenario: 5a. **Incorrect e-mail or password** 1. The system detects an incorrect email or password. 2. The system shows an "incorrect email or password" error, return to MSS at step 1.

2.2. Doctor

The following actions can be undertaken by the physician to ensure continuous monitoring of patients suffering from sleep disorders.

2.2.1. Doctor views data

This activity describes the process by which the doctor can access, analyze, and comment on a patient's recorded data using the system.

Title: Doctor views data
Pre-condition: Patient data has been previously recorded and stored in the database.

Post-condition: The selected patient's data has been displayed.
Main success scenario 1. The doctor opens the patient selection window 2. The doctor select the patient 3. The software opens the patient personal page 4. The doctor press the "View Data" button 5. The software opens the "Data Selection window" 6. The Doctor select the time window and the type of data that he wants to visualize 7. The software displays the value 8. The software check the values for abnormal data 9. The doctor may insert a comment 10. The software saves the comment
Alternative scenario 2a. **No Patient Selected** 1. The doctor does not select any patient. 2. The system prompts an error message requesting patient selection. 4a. **Data Not Available** 1. The doctor clicks "View Data". 2. The system notifies the doctor that no data is available. 6a**. Invalid Time Window Selected** 1. The doctor selects an invalid or empty time window. 2. The system notifies the doctor and requests a new selection. 8a. Abnormal values Detected 1. The software marks the abnormal values

2.2.2. Doctor contacts patient

Title: Doctor contacts patient
Pre-condition: The patient's contact information is available and up-to-date.
Post-condition: The message has been successfully sent to the patient.
Main success scenario 1. Open the patient selection window 2. The software displays the patient's list 3. The doctor select a patient 4. Open the window of the chosen patient 5. The Doctor clicks on the message button 6. The software opens the message window 7. The doctor writes the message 8. The software sends the message

Alternative scenario 5b. The doctor wants to call the patient 1. The doctor clicks on the patient's phone number 2. The doctor calls the patient and the activity ends

2.2.3. Prescription managment

This activity describes the process through which the doctor reviews a patient's prescription, verifies the remaining medication, checks for possible misuse, and, if necessary, sends a new prescription. The patient is then required to confirm receipt of the updated prescription.

Title: Prescription managment
Pre-condition: The patient has an active medical prescription associated with their profile.
Post-condition: The patient has received and confirmed the new prescription.
Main success scenario 1. The doctor opens the patient's profile. 2. The doctor checks the current prescription status. 3. The doctor verifies the remaining quantity of medication. 4. The doctor evaluates the patient's usage history to detect any potential misuse. 5. If the prescription is close to running out or misuse is suspected, the doctor decides whether to contact the patient. 6. The doctor sends the new prescription to the patient through the system. 7. The patient receives the new prescription. 8. The patient confirms receipt of the prescription.
Alternative scenario 4a. **Misuse Detected** 1. If the doctor detects irregularities or potential misuse in the medication usage, the doctor may contact the patient to schedule a follow-up appointment or adjust the treatment plan. 6a. **Prescription Sending Failure** 1. If the system encounters an error while sending the prescription, it notifies the doctor. 2. The doctor may retry the operation or choose an alternative communication channel. 7a. **Prescription Not Received** 1. If the patient does not receive the prescription within a reasonable time frame, the patient notifies the doctor. 2. The doctor verifies the system status and resends the prescription if necessary. 8a. **No Confirmation from Patient** 1. If the patient does not confirm receipt of the prescription, the system sends an automatic reminder. 2. If no response is received after the reminder, the doctor may contact the patient directly.

2.2.4. Specialist visit request

This activity describes the process through which the doctor, upon detecting significant anomalies in the data collected by the patient's device, evaluates the need for a specialist consultation. If the doctor suspects a worsening of the sleep disorder or determines that the current therapy might need revision, a specialist visit is scheduled. The patient is required to confirm the appointment or, if unavailable, cancel and reschedule it.

Title: Specialist visit request
Pre-condition: The doctor has reviewed the patient's data and identified strong anomalies.
Post-condition: A specialist visit has been scheduled and confirmed by the patient.
Main success scenario 1. The doctor reviews the patient's data collected by the device. 2. The doctor detects significant anomalies or patterns suggesting a potential aggravation of the sleep disorder. 3. The doctor evaluates whether a specialist visit is necessary. 4. If a specialist visit is deemed necessary, the doctor schedules the appointment through the system. 5. The patient receives the visit notification. 6. The patient confirms the appointment. 7. The system records the confirmation and updates the patient's agenda.
Alternative scenario 4a. **Specialist Unavailable for the Suggested Timeframe** 1. If no specialist is available for the desired time slot, the system suggests the next available dates. 2. The doctor selects a new date and sends the updated appointment to the patient. 5a. **Appointment Notification Failure** 1. If the system fails to deliver the appointment notification to the patient, an error message is shown to the doctor. 2. The doctor can attempt to resend the notification or contact the patient through alternative channels. 6a. **Patient Unable to Attend the Scheduled Visit** 1. The patient cancels the appointment via the system. 2. The patient selects a new available date and reschedules the specialist visit. 3. The system updates the appointment and notifies both the patient and the doctor of the new date.

2.3. Patient

The following activities describe the interactions and actions performed by the patient within the system. These include reporting issues related to the monitoring device, managing communications with the doctor, confirming the reception of medical prescriptions and devices, and handling appointments. The patient actively collaborates with both the healthcare provider and the system to ensure the continuity and effectiveness of the monitoring and treatment process.

2.3.1. Device Replacement Request

This activity describes the process in which the patient notices that the device is no longer functioning properly — either due to accidental damage or technical failure — and requests a replacement through the system. Once the new device is delivered, the patient is required to confirm its reception via the app.

Title: Device Replacement Request
Pre-condition: The patient is registered in the system and has been assigned a monitoring device.
Post-condition: The patient has received and confirmed the new device.
Main success scenario 1. The patient realizes the device is not working properly. 2. The patient accesses the system and requests a device replacement. 3. The system registers the replacement request and forwards it to the responsible service. 4. A new device is prepared and delivered to the patient. 5. The patient receives the new device. 6. The patient confirms the reception of the device through the app. 7. The system updates the patient's profile with the new device information.
Alternative scenario 5a. **Delivery Delays or Issues** 1. If the device is not delivered within the expected time frame, the patient can notify the support service through the app. 2. The support team verifies the delivery status and provides an update or arranges a new shipment if necessary. 6a. **Device Not Functioning After Replacement** 1. If the patient detects that the replacement device is not working correctly after reception, the patient can submit a malfunction report through the app. 2. The system logs the report and initiates a technical check or dispatches a new replacement if needed.

2.4. Device

The following activities reguard the device worn by the patient in order to register the sleeping activity-

2.4.1. Device functionality

This activity describes the core functionality of the sensor device, which is used by the patient to monitor specific health parameters. The process includes device initialization, automatic self-check for correct placement, measurement acquisition, data storage, and real-time feedback to the patient. The system also performs periodic data analysis to provide summarized insights.

Title: Device funtionality
Pre-condition: The device is fully charged and operational and the patient has been instructed on how to correctly wear the device.
Post-condition: Measurement data has been successfully collected and stored.
Main success scenario 1. The patient wears the sensor device. 2. The device sensors check whether the positioning is correct. 3. If the positioning is correct, the process continues. 4. The device starts recording measurements. 5. The measurements are saved in the system's database. 6. The recorded data is displayed to the patient via the device screen. 7. At regular intervals, the system performs an overall analysis of the collected data.
Alternative scenario 2a. **Incorrect Positioning Detected** 1. If the device detects that the positioning is not correct, it alerts the patient. 2b. **Continuous Position Check** 1. The device continuously checks the sensor placement until the positioning is corrected.

3. Use case diagrams

4.Activity diagrams

5. Class diagrams

6. Database design

7. System implementation

8. Conclusions